Run report: E3 sessions, 2026-10-19T23:34Z to 2026-10-20T05:35Z

Observations only. The interpretation is in `../../E3.md`.

item	value
protocol	`E3-PROTOCOL.md`, frozen in commit `ede431274` before these sessions
specs	`e3/specs/` (`e3-ctl`, `e3-c1`, `e3-c2`, `e3-c3`)
command	`ecbench run --allow-busy --lock $TMPDIR/crypto-bench.lock --spec e3/specs/<c>.json --out e3/sessions/<c>`, one curve after another; build at `ede431274`
host	macOS (Apple silicon), shared, no CPU topology, so `--cpus none`. Every record is at isolation level L0. Other jobs ran concurrently. Wall time is not used
sessions	`e3-ctl` ECBS1h28077f2dbe1c (23:34:15–23:34:19Z), `e3-c1` ECBS1h642a1a44f9f9 (23:34:19Z–01:34:48Z), `e3-c2` ECBS1hd4736f2baf84 (01:34:48–03:32:32Z), `e3-c3` ECBS1h2a56f971e518 (03:32:32–05:34:44Z)
arms	per curve: `rho-neg`; per t: `sem-chain-t{t}`, `sem-full-t{t}` ($t \le 4$), `sem-grid-t{t}`, `kl-mitm3-t{t}`; control: t = 2 only, `sem-chain` uncapped
workloads	8 `planted_shared` targets per curve, target seed 20261019, algorithm seed 20261019, 1 round, no warm-up
requested / completed	416 / 416 runs. Every full-solve run verified: 248, comprising 72 per 24-bit curve and 32 on the control. Every capped F4 run `exhausted` at its `max_trials` as specified: 168, 56 per 24-bit curve. No failures, timeouts or errors
raw records	`e3/sessions/<c>/records.jsonl` and `session.json`

Measured values (operation counts; S = GAE/√r at w = 0, mean of 8 runs)

curve	rho-neg	grid t2 / t3 / t4 / t5	known-log MITM t2 / t3 / t4 / t5
e3-c1	1.079	389.3 / 128.9 / 31.68 / 6.37	1111 / 246.2 / 48.69 / 9.30
e3-c2	1.329	316.9 / 134.0 / 26.93 / 7.30	469.4 / 243.0 / 56.79 / 16.74
e3-c3	1.174	1845 / 277.0 / 33.81 / 8.24	2553 / 572.3 / 53.55 / 27.36
control (12-bit)	2.728	18.38 (t2)	14.34 (t2); `sem-chain-t2` uncapped 18.38

- **F4 multiply-adds per call** (capped records, all curves): chain $3.908 \cdot 10^{5}$, $1.587 \cdot 10^{8}$, $1.060 \cdot 10^{10}$ and $3.186 \cdot 10^{12}$ at t = 2…5; full $2.243 \cdot 10^{5}$, $1.260 \cdot 10^{7}$ and $1.683 \cdot 10^{10}$ at t = 2…4.
- **Grid multiplications per call:** 244, 1,224, 7,440 and 51,232.
- **Cross-check:** `pkm_crosscheck_disagreements` = 0 over 24,502 capped F4 calls, all of them refutations.
- **Pricing:** the weights 0.1 and 1, the intervals and the derived ratios are in `e3/summary.json`, produced by `examples/generator_effect_e3_summary.rs`.

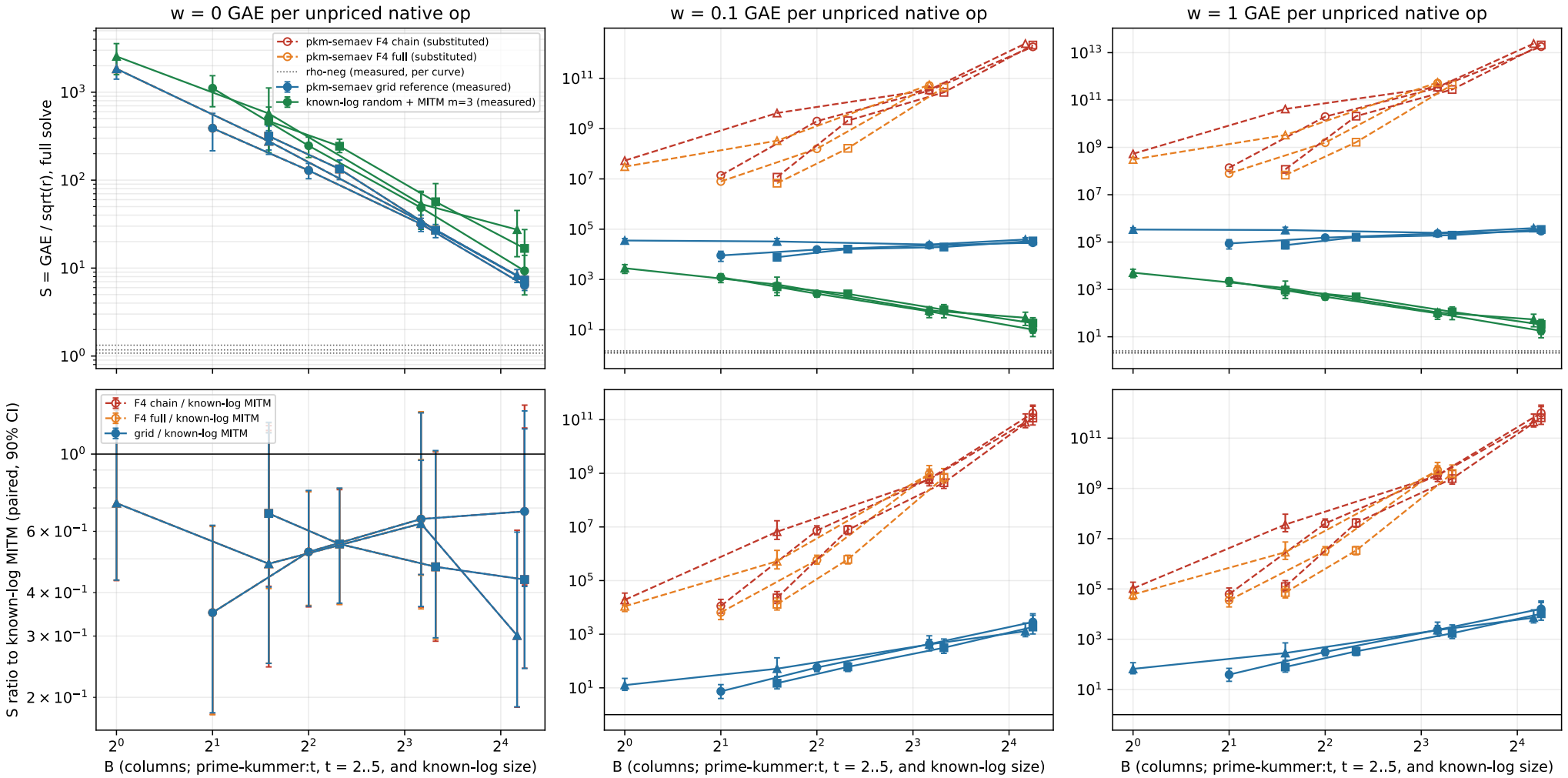


Figure 1: S against B at the three band weights

S against B at w = 0, 0.1 and 1, sessions as above. F4 arms dashed and substituted (see E3.md). 8 targets per point, 90% bootstrap intervals.